AGE-Sampler: Adaptive-granularity Graph-enhanced Event Sampling for Long-Video Understanding

Yanzi Wang[1] **Tao Dai**[2*] **Yaohua Zha**[1] **Shu-Tao Xia**[1]
[1]Tsinghua Shenzhen International Graduate School, Tsinghua University
[2]College of Computer Science and Software Engineering, Shenzhen University
wangyanzi0314@gmail.com daitao.edu@gmail.com
zyh1614399882@gmail.com xiast@sz.tsinghua.edu.cn

Abstract

Informative keyframe selection is critical for applying large vision-language models (LVLMs) to long-form video understanding under limited frame budgets. However, existing methods heavily rely on query-frame similarity, which can yield coarse boundaries and miss latent contextual events that are answer-relevant but not explicitly mentioned in the query. To address this issue, we propose **AGE-Sampler**, a training-free **A**daptive-granularity **G**raph-enhanced **E**vent sampling framework for long-video understanding. AGE-Sampler jointly exploits low-level visual transition cues and query-relevance signals to construct query-adaptive events, thus preserving fine-grained structures in relevant regions while merging low-relevance regions into compact intervals. To recover answer-bearing contextual evidence, we further develop a fine-grained event graph and propagate importance from high-confidence query-relevant anchors via Personalized PageRank, followed by event-level budget allocation and representative keyframe selection. Extensive experiments with different LVLM backbones show that our AGE-Sampler consistently improves long-video understanding performance, achieving average absolute gains of **+6.2% on LongVideoBench**, **+8.8% on MLVU**, and **+3.8% on Video-MME**, while remaining lightweight and plug-and-play. Project Page: https://yanzi-wang.github.io/age-sampler

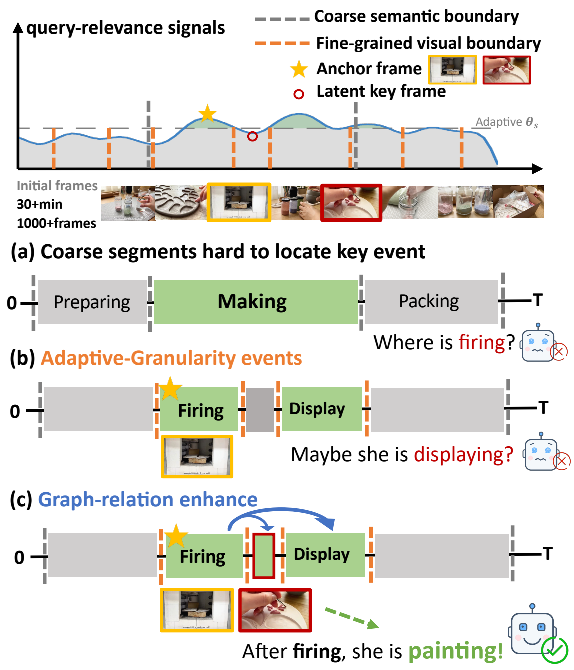

Figure 1: Motivation. For the query *"What did the blogger do next after **firing** the **ceramics**?"*: (a) coarse query–frame similarity leads to over-large segments where the key event is lost; (b) fine-grained visual scene cuts help, but pure similarity matching still misses inherently relevant frames (e.g. ***painting***); (c) our method propagates importance over an event graph to recover latent contextual events.

1 Introduction

Recent advances in large-scale Vision-Language Models (VLMs) (Bai et al., 2025; Li et al., 2025; Zhu et al., 2025) have extended multimodal reasoning from static images to videos, enabling stronger understanding of temporal dynamics and event-level semantics. As these models scale, efficient visual processing has become increasingly important for handling diverse and long-form inputs (Zha et al., 2025; Gao et al., 2024). Yet long videos pose a fundamental input bottleneck: the sheer number of frames, many of them redundant, conflicts with limited computational budgets and fixed-size context windows. Therefore, selecting the most informative visual evidence under a constrained frame budget has become a central problem in long-video understanding.

Early methods usually rank and filter video frames independently based on frame-level query relevance scores (Nasreen and Shobha, 2013). However, such methods ignore the inherent temporal continuity and event structure of videos, often

*Corresponding author.

leading to temporally discontinuous sampling results that fail to fully capture the narrative process of long videos. Therefore, recent studies (Chen et al., 2026; Tang et al., 2025; Zou et al., 2026) have increasingly focused on event-level organization and structured modeling to identify query-grounded events that carry the supporting evidence.

Despite the great progress, current methods rely heavily on query-frame matching signals for event extraction, scoring, and frame selection, thus suffering from two main limitations. **First, they suffer from coarse event granularity.** Query–frame similarity is often noisy, so prior work smooths or denoises it (Chen et al., 2026), blurring boundaries and limiting fine-grained segmentation needed for detail-oriented queries. As illustrated in Figure 1(a), for the question *"What did the blogger do next after firing the ceramics?"*, the queried event is extremely brief (the *firing* clip lasts only $\sim$3 s); overly coarse event boundaries may fail to isolate such critical moments. **Second, latent contextual events are hard to match.** Answering complex video questions often requires reasoning about both temporal order and semantic relationships among events, not simply retrieving frames that match query keywords. Similarity-based approaches tend to overemphasize salient queried entities (e.g., *ceramics* or *fire*) and may overlook later, answer-critical moments. As illustrated in Figure 1(b), the *painting* event obtains a low CLIP score (Li et al., 2022, 2023; Radford et al., 2021; Zhai et al., 2023) and is consequently excluded by purely similarity-driven selection.

To address these limitations, we propose **AGE-Sampler**, an **A**daptive-granularity **G**raph-enhanced **E**vent sampling framework for long-video understanding. As shown in Figure 2, AGE-Sampler first fuses low-level visual transition cues with query-relevance signals to form query-adaptive events: it retains fine-grained boundaries in highly relevant regions while collapsing low-relevance segments into compact intervals. It then constructs a fine-grained event graph and propagates importance from high-confidence, query-relevant anchors using Personalized PageRank, amplifying latent contextual events that are only weakly captured by direct similarity. Extensive experiments show consistent gains in LVLM performance across multiple long-video understanding benchmarks. For instance, with InternVL3-8B (Zhu et al., 2025) under a 32-frame input budget, AGE-Sampler improves accuracy by **5.6%** on LongVideoBench (Wu et al., 2024), **6.8%** on MLVU(Zhou et al., 2025), and **2.5%** on Video-MME(Fu et al., 2025), outperforming existing frame-sampling baselines.

The main contributions are summarized as follows:

- We propose **AGE-Sampler**, a training-free *adaptive-granularity graph-enhanced event sampling* framework for long-video understanding that builds query-adaptive event representations by jointly leveraging visual transition cues and query-relevance signals.

- We introduce a *graph-enhanced contextual importance propagation* mechanism that transfers importance from query-relevant anchor events to latent contextual events, mitigating evidence omission inherent to direct similarity matching.

- Extensive experiments on multiple long-video question answering benchmarks demonstrate that AGE-Sampler consistently boosts a range of video understanding models, confirming its effectiveness, generality, and plug-and-play nature.

2 Related Work

Long-Video Understanding with LVLMs. Recent Large Vision-Language Models (LVLMs) (Bai et al., 2025; Li et al., 2025; Zhu et al., 2025) have substantially advanced video understanding, but their fixed context windows make hour-scale videos challenging. One line of work extends model capacity through architectural changes such as spatiotemporal compression (Shen et al., 2024) or extra-long context mechanisms (Shu et al., 2025); another aggressively abstracts the visual stream into trees (Wang et al., 2025), retrieved snippets (Luo et al., 2025b), or pruned token sets (Luo et al., 2025a). While effective, these strategies introduce non-trivial training or inference overhead, which motivates lightweight, plug-and-play *frame selection* as a complementary pre-processing step.

Frame Selection for Long-Video Understanding. Early keyframe extractors rely on low-level, query-agnostic cues such as shot boundaries (Nasreen and Shobha, 2013). Modern approaches are query-aware and broadly fall into two camps. *Training-based* selectors learn a sampling policy end-to-end (Yu et al., 2025; Fang et al., 2026), but

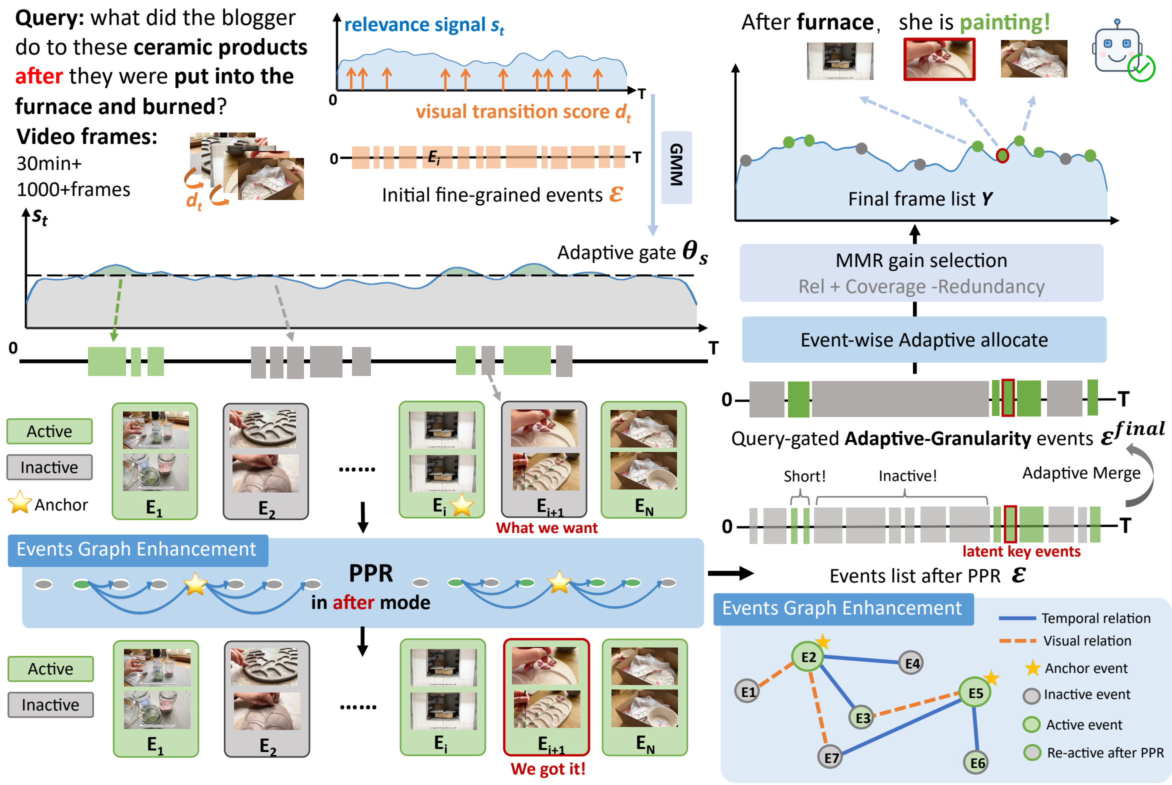

Figure 2: Overview of **AGE-Sampler**. Given a frame sequence $\mathcal{F} = \{f_t\}_{t=1}^{T}$ and a query q, we compute frame-query relevance s_t and visual transition d_t to construct query-gated fine-grained event candidates $\mathcal{E}$. An event graph is then built over $\mathcal{E}$, where Personalized PageRank (PPR) propagates importance from query-relevant anchors to latent contextual events. The graph-refined event scores guide adaptive merging, producing the final query-gated adaptive-granularity event set $\mathcal{E}^{\text{final}}$. Based on $\mathcal{E}^{\text{final}}$, AGE-Sampler allocates the event-level frame budget $\{k_i\}$ and samples representative keyframes Y for LVLM inference.

they are data-hungry and tied to a specific backbone. *Training-free* selectors are more flexible and are dominated by two paradigms. The first consists of lightweight *similarity-driven* samplers that score each frame with a contrastive vision-language model and then post-process the score sequence: BOLT (Liu et al., 2025) applies inverse transform sampling, MDP^3 (Sun et al., 2025) casts selection as a list-wise Markov decision process, Q-Frame (Zhang et al., 2025b) additionally adapts the spatial resolution of each selected frame, AKS (Tang et al., 2025) recursively splits the video and judges each half, WFS (Chen et al., 2026) detects semantic boundaries in the wavelet domain, and T* (Ye et al., 2025) performs object-grounded temporal search. The second consists of *VLM-assisted* samplers that invoke a powerful (often proprietary) VLM to caption, reason over, or iteratively refine the candidate frames, such as VideoAgent (Wang et al., 2024), MVU (Ranasinghe et al., 2025), LLoVi (Zhang et al., 2024), the hierarchical tree representation of VideoTree (Wang et al., 2025), and the iterative reasoning-based A.I.R. (Zou et al., 2026).

While similarity-driven methods are efficient, they score frames in isolation and therefore ignore event-level structure; VLM-assisted methods recover more context but incur substantial extra inference cost, which we quantify against A.I.R. in Sec. 4.2. Our AGE-Sampler keeps the lightweight training-free nature of the former while introducing event-level reasoning: it constructs query-adaptive events and propagates importance over an event graph to recover latent contextual evidence without additional VLM inference.

3 Method

We study video question answering (VidQA), where a video-query pair (V, q) is given and the model is required to answer the query based on visual evidence in the video. Since long videos contain massive redundant frames, directly feeding all frames into an LVLM is usually infeasible. We therefore uniformly decode the video into a candidate frame sequence $\mathcal{F} = \{f_t\}_{t=1}^{T}$, and aim to select a compact keyframe set $Y \subset \mathcal{F}$ with $|Y| = K$ and $K \ll T$.

An effective keyframe set should not only capture query-relevant evidence, but also preserve the temporal event structure required for reasoning. Therefore, we formulate long-video frame selection as a query-conditioned event structure refinement problem, rather than independently ranking frames by relevance scores.

This formulation raises three key challenges: how to construct an event structure with appropriate granularity, how to uncover latent contextual evidence beyond direct frame-level matching, and how to allocate a limited frame budget over the resulting event structure.

To this end, we propose **AGE-Sampler**, a training-free **A**daptive-granularity **G**raph-enhanced **E**vent sampling framework, as illustrated in Fig. 2. AGE-Sampler consists of three core modules. First, query-gated event segmentation computes frame-query relevance s_t and visual transition signals d_t to construct fine-grained event candidates with query-aware relevance priors. Second, pre-merge graph-enhanced event reweighting builds an event graph over these candidates and propagates importance from high-confidence query-relevant anchors via Personalized PageRank, so as to recover latent contextual events that are weakly captured by direct similarity. Third, adaptive event refinement and hierarchical sampling merge low-relevance regions into compact intervals, allocate the frame budget across the refined event structure, and select representative keyframes within each event.

The following sections detail these modules. Sec. 3.1 introduces query-gated event segmentation. Sec. 3.2 describes pre-merge graph-enhanced event reweighting. Sec. 3.3 presents adaptive event merging, event-level budget allocation, and representative keyframe selection.

3.1 Query-Gated Adaptive-Granularity Event Segmentation

Given a sampled frame sequence $\{f_t\}_{t=1}^{T}$ and a query q, we compute two complementary temporal signals: a frame-query relevance score and a frame-frame visual transition score,

$$s_t = \mathrm{ITM}(f_t, q), \qquad d_t = H(f_{t-1}, f_t), \tag{1}$$

where $\mathrm{ITM}(\cdot)$ denotes a pretrained image-text matching model, instantiated with BLIP-2 (Li et al., 2023) in our implementation (Sec. 4.1), and $H(\cdot)$ measures the low-level appearance discontinuity between adjacent frames. The former provides query-aware semantic evidence, while the latter captures query-agnostic visual changes for detecting fine-grained shot boundaries. Details of $H(\cdot)$ are provided in Appendix A.

We first estimate a dynamic relevance threshold from the semantic score sequence $s_{1:T}$. Specifically, we assume that the frame-level relevance scores are drawn from a mixture of two latent distributions, corresponding to query-relevant frames and background frames. We fit a two-component Gaussian Mixture Model (Huang and Chau, 2008) and we define the adaptive relevance threshold as

$$\theta_s = \max(\mu_1, \mu_2) - \gamma \max(\sigma_1, \sigma_2), \tag{2}$$

where μ_k and σ_k denote the mean and standard deviation of the k-th Gaussian component, and γ controls the strictness of the query gate. This threshold adaptively separates query-relevant regions from background regions according to the score distribution of each video-query pair.

Meanwhile, the visual transition signal $d_{1:T}$ is used to produce an initial fine-grained event set:

$$\mathcal{E} = \{E_i = [a_i, b_i)\}_{i=1}^{M}. \tag{3}$$

These atomic events preserve local visual boundaries, but they are generated from query-agnostic visual changes and may contain many irrelevant fragments in long videos.

Therefore, we further assign each event a query-aware importance score. For an event segment E, let $\bar{s}_E$ denote its average relevance score, $s_E^{\max}$ denote its peak relevance score, and $\ell_E = |E|/T$ denote its normalized duration. We define the event-level importance operator as

$$\Phi(E; s) = w_\mu \bar{s}_E + w_M s_E^{\max} + w_d \ell_E. \tag{4}$$

The three terms respectively measure the average relevance, peak relevance, and temporal coverage of the event. For each initial event E_i, its raw query-aware importance is computed as $u_i = \Phi(E_i; s)$.

The dynamic threshold θ_s is then used in Sec. 3.3 to partition the initial events into a high-relevance subset $\mathcal{E}^+$ and a low-relevance subset $\mathcal{E}^-$. Events in $\mathcal{E}^-$ are treated as inactive background candidates, and are more likely to be merged into compact coarse segments during the subsequent event refinement stage.

3.2 Graph-Enhanced Latent Event Reweighting

Event graph construction. Before merging the initial event proposals, we perform event-level relation modeling to refine their importance scores. We construct an event graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{A}), \quad \mathcal{V} = \mathcal{E}, \tag{5}$$

where each node corresponds to an event candidate and $\mathcal{A} \in \mathbb{R}^{M \times M}$ is the weighted adjacency matrix.

The graph contains two types of relations: temporal relation and visual relation. The temporal relation models query-conditioned ordering dependencies among events, while the visual relation captures appearance similarity and local continuity in the visual feature space.

Temporal relation. Let $c_i = (a_i + b_i)/2$ denote the midpoint of event interval. We use a lightweight temporal cue parser to map the query into a coarse temporal mode $m(q) \in \{\texttt{after}, \texttt{before}, \texttt{generic}\}$. The temporal adjacency is defined as

$$A_{ij}^{\text{temp}} = D_{ij}^{q} \exp\left(-\frac{|c_i - c_j|}{\tau_t}\right), \tag{6}$$

where

$$D_{ij}^{q} = \begin{cases} \mathbb{I}(c_j > c_i), & m(q) = \texttt{after}, \\ \mathbb{I}(c_j < c_i), & m(q) = \texttt{before}, \\ 0, & m(q) = \texttt{generic}. \end{cases} \tag{7}$$

Thus, temporal propagation follows the direction indicated by the query.

Visual relation. Besides temporal dependency, two events may also be related through visual semantics or local visual continuity. For each event E_i, we obtain an event-level visual representation $\mathbf{h}_i$ by average pooling the frame features within its interval, we then define a visual adjacency to connect events with similar appearances or local continuity:

$$A_{ij}^{\text{vis}} = A_{ij}^{\text{sem}} + \lambda_c A_{ij}^{\text{cont}}, \tag{8}$$

where A_{ij}^{sem} is computed from the cosine similarity between event representations $\mathbf{h}_i$ and $\mathbf{h}_j$, and A_{ij}^{cont} measures the visual continuity between adjacent events. The details of these two terms are provided in Appendix B.

Final edge weight. The final adjacency matrix is

$$\mathcal{A} = \lambda_t^q A^{\text{temp}} + \lambda_v^q A^{\text{vis}} + \lambda_0 \mathbf{I}, \tag{9}$$

where the self-loop term keeps the propagation conservative.

Graph-based event reweighting. We select a small set of high-confidence anchor events according to their direct query relevance and query-conditioned temporal prior. These anchor events serve as source nodes for importance propagation. After row-normalizing $\mathcal{A}$ into a transition matrix P, we apply Personalized PageRank:

$$\mathbf{p}^{(k+1)} = (1-\alpha)\mathbf{y} + \alpha P^{\top} \mathbf{p}^{(k)}, \tag{10}$$

where $\mathbf{y}$ is the personalized distribution concentrated on the anchor events and α is the damping factor. The resulting score p_i measures how strongly event E_i is connected to the anchor events through query-conditioned temporal and visual relations.

Finally, we fuse the propagated graph score with the original event importance:

$$\widetilde{u}_i = (1-\eta) u_i + \eta p_i, \tag{11}$$

where u_i denotes the original event importance and η controls the strength of graph enhancement. The reweighted event importance $\widetilde{u}_i$ is used in the subsequent adaptive merging step. This allows latent key events to receive higher importance scores, thereby increasing their preservation priority during the subsequent merging and sampling process.

3.3 Adaptive Event Merging and Event-Structured Frame Sampling

Event-level Budget Allocation. After graph-enhanced reweighting, we further refine the over-complete initial events through bottom-up merging. As described in Sec. 3.1, the adaptive threshold θ_s separates events into high- and low-relevance groups. Here, we perform merging based on the graph-reweighted event score $\widetilde{u}_i$.

For two adjacent events E_i and E_j, we define

$$\ell_{ij} = \max\{|E_i|, |E_j|\}, \qquad u_{ij} = \max\{\widetilde{u}_i, \widetilde{u}_j\}. \tag{12}$$

The merge cost is then defined as

$$C(i,j) = \begin{cases} -1, & \ell_{ij} < L_s, \\ u_{ij}, & u_{ij} < \theta_s, \\ +\infty, & \text{otherwise}. \end{cases} \tag{13}$$

Here, L_s denotes the short-event length threshold. The first case handles very short neighboring events, and the second case merges neighboring low-relevance events into coarser background segments. In contrast, high-relevance regions preserve their fine-grained event structure, so their sampling budget will not be diluted by adjacent low-relevance regions in the subsequent allocation stage. We iteratively merge the adjacent pair with the lowest cost until reaching the target number of events or no valid pair remains. The resulting refined event set is denoted as

$$\mathcal{E}^{final} = \{E_i^{final} = [a_i, b_i)\}_{i=1}^{L}. \quad (14)$$

Given the refined event structure and the graph-reweighted event score $\widetilde{u}_i$, we allocate the total frame budget K at the event level.

$$k_i = \mathrm{round}\left(K \cdot \frac{\exp(\widetilde{u}_i/\tau_b)}{\sum_{j=1}^{L} \exp(\widetilde{u}_j/\tau_b)}\right), \quad (15)$$

where τ_b controls the smoothness of budget allocation.

Intra-event Frame Sampling. For simplicity, we denote each refined event E_i^{final} as E_i in this paragraph. Within each event, we select representative frames using a localized MMR criterion. Let S_i^{r-1} be the frames selected from E_i before the r-th step. For each unselected frame $t \in E_i \setminus S_i^{r-1}$, its selection gain is defined as

$$g(t) = \lambda_r \,\mathrm{rel}(t) + \lambda_c \,\mathrm{cov}(t) - \lambda_d \,\mathrm{red}(t). \quad (16)$$

Here, $\mathrm{rel}(t)$ measures query relevance, $\mathrm{cov}(t)$ encourages temporal coverage within E_i, and $\mathrm{red}(t)$ penalizes visual redundancy with the selected frames S_i^{r-1}. Detailed definitions are given in Appendix C. We greedily select the frame with the largest gain until k_i frames are chosen from each event. Let Y_i be the selected frame set from event E_i after k_i steps. The final keyframe set is

$$Y = \bigcup_{i=1}^{L} Y_i, \qquad |Y| = K. \quad (17)$$

4 Experiments

4.1 Experimental Setup

Backbone models and benchmarks. We evaluate our method on top of four widely used large vision–language models that span different architectures and training recipes: LLaVA-OneVision-7B(Li et al., 2025), InternVL3-8B(Zhu et al., 2025), Qwen2.5-VL-7B(Bai et al., 2025) and MiniCPM-V2.6(Yao et al., 2024). To study how the proposed sampler interacts with model capacity, we further conduct a multi-scale ablation on the Qwen2.5-VL family, covering the 3B, 7B, 32B, and 72B variants. We benchmark all methods on three long-video question answering datasets: LongVideoBench(Wu et al., 2024) (LVB; the 1,337-pair validation set, average video length 12 min), MLVU(Zhou et al., 2025) (2,174 questions across 7 task categories, average 11 min), and VideoMME(Fu et al., 2025) (900 videos with 2,700 QA pairs, average 17 min). Subtitles are not used in any experiment.

Baselines. We compare our AGE-Sampler with a set of representative keyframe selection methods, ranging from similarity-driven samplers such as WFS(Chen et al., 2026) to LVLM-assisted samplers such as A.I.R.(Zou et al., 2026), specifically: BOLT (Liu et al., 2025), AKS (Tang et al., 2025), MDP[3] (Sun et al., 2025), WFS (Chen et al., 2026), and A.I.R. (Zou et al., 2026). Methods with publicly available implementations (†) are re-run in our environment under a single protocol that fixes every dimension known to affect frame selection: 1 fps candidate extraction, one prompt per benchmark shared by all methods, the same LVLM backbone, the same frame budget K, and the same `lmms-eval` toolkit. All † rows within a model group therefore share an identical uniform-sampling base and directly comparable Δ values. Methods without released code ($*$) are quoted from their original papers under the closest available settings, each carrying the base reported by its own authors, so comparing their Δ against the † rows is indicative rather than controlled. Table 1 marks the two groups accordingly, and Sec. 4.2 adds a fully controlled head-to-head comparison against A.I.R.

Implementation details. Candidate frames are extracted from each video at 1 fps, and frame–query matching scores are produced by BLIP-2 (Li et al., 2023). The relaxation factor for the adaptive relevance threshold is set to $\gamma = 0.7$. All experiments sweep the frame budget over $K \in \{8, 16, 32\}$ on NVIDIA H100 GPUs with 80 GB of memory. We report the benchmarks' accuracy as measured by the `lmms-eval` toolkit (Zhang et al., 2025a).

Model	Method	Size	Frames	LongVideoBench			MLVU			VideoMME		
				Base	+Method	Δ	Base	+Method	Δ	Base	+Method	Δ
LLaVA-OV	BOLT*	7B	32	56.4	59.6	+3.2	63.2	66.8	+3.6	58.5	59.9	+1.4
	AKS*			54.8	59.3	<u>+4.5</u>	–	–	–	56.5	58.4	+1.9
	A.I.R*,§			58.5	<u>60.7</u>	+2.2	62.4	**69.3**	**+6.9**	58.3	<u>61.4</u>	<u>+3.1</u>
	MDP3†			56.3	59.5	+3.2	62.1	68.2	+6.1	58.0	59.9	+1.9
	Ours†			56.3	**61.8**	**+5.5**	62.1	<u>68.9</u>	<u>+6.8</u>	58.0	**61.7**	**+3.7**
Qwen2.5-VL	AKS*	7B	32	58.9	63.2	+4.3	59.7	67.2	+7.5	61.2	64.0	+2.8
	A.I.R*,§			58.1	61.4	+3.3	59.3	67.5	+8.2	60.8	**65.0**	**+4.2**
	MDP3†			58.5	59.5	+1.0	59.4	66.1	+6.7	61.1	63.2	+2.1
	WFS†			58.5	<u>64.4</u>	<u>+5.9</u>	59.4	<u>69.5</u>	<u>+10.1</u>	61.1	64.2	+3.1
	Ours†			58.5	**64.9**	**+6.4**	59.4	**69.9**	**+10.5**	61.1	<u>64.5</u>	<u>+3.4</u>
MiniCPM-V2.6	MDP3†	7B	8	51.6	57.3	+5.7	51.6	62.2	+10.6	52.8	<u>58.0</u>	<u>+5.2</u>
	WFS†			51.6	<u>58.6</u>	<u>+7.0</u>	51.6	**62.6**	**+11.0**	52.8	57.9	+5.1
	Ours†			51.6	**59.0**	**+7.4**	51.6	<u>62.5</u>	<u>+10.9</u>	52.8	**58.5**	**+5.7**
InternVL-3	AKS*	8B	32	58.5	61.5	+3.0	68.4	74.2	+5.8	65.6	66.3	+0.7
	A.I.R*,§			58.3	<u>62.8</u>	<u>+4.5</u>	68.4	<u>74.5</u>	<u>+6.1</u>	65.6	**68.2**	**+2.6**
	MDP3†			58.3	60.4	+2.1	68.3	73.6	+5.3	65.1	66.8	+1.7
	WFS†			58.3	62.5	+4.2	68.3	74.3	+6.0	65.1	67.1	+2.0
	Ours†			58.3	**63.9**	**+5.6**	68.3	**75.1**	**+6.8**	65.1	<u>67.6</u>	<u>+2.5</u>

Table 1: Performance comparison of different LVLMs and frame selection methods on LongVideoBench, MLVU, and VideoMME. We report accuracy scores (%). "Base" denotes uniform frame sampling, "+Method" denotes applying the corresponding key-frame selection method, and "Δ" denotes the absolute gain over the base of that row. Rows marked † are reproduced by us under a single shared protocol (Sec. 4.1); within each model group they share an identical base, so their Δ values are directly comparable. Rows marked * are quoted from the original papers and each carries the uniform-sampling base reported there, so comparing their Δ against the † rows is indicative rather than controlled. § marks an LVLM-dependent iterative method. **Bold** indicates the best performance and <u>underlined</u> indicates the second-best performance in each model–benchmark group.

4.2 Main Results

Table 1 reports the comparison across LVLM backbones, benchmarks, and frame budgets. Three observations stand out.

Strong generality and plug-and-play behaviour. Our AGE-Sampler is backbone-agnostic and requires no additional training, yet it achieves consistent gains across all four LVLM backbones and three benchmarks. The improvements are particularly pronounced on MLVU, with absolute accuracy gains of $+10.5\%$ for Qwen2.5-VL and $+10.9\%$ for MiniCPM-V2.6. AGE-Sampler also achieves a $+7.4\%$ gain on LongVideoBench and a $+5.7\%$ gain on VideoMME with MiniCPM-V2.6, showing its robustness across benchmarks with different video-length distributions.

Competitive accuracy at substantially lower sampling cost. As shown in Table 1, AGE-Sampler already attains competitive or superior accuracy across model–benchmark combinations; here we focus on how cheaply this is obtained. Our main competitor, the LVLM-assisted A.I.R. baseline, invokes an LVLM iteratively to reason about and select frames before the final VidQA inference, whereas AGE-Sampler only uses cached matching signals and lightweight graph operations. To quantify what this design choice buys, we compare the two methods head to head on VideoMME with

Method	Sampling (s/QA)	Online E2E (s/QA)	LVLM calls/QA	Sampling GPU (GiB)
A.I.R.	15.880	19.101	7.12	16.44–22.72
Ours	**0.426**	**3.987**	**1.00**	≈0
Advantage	37.3×	4.8×	7.12×	—

Table 2: Controlled efficiency comparison against A.I.R. on VideoMME with InternVL3-8B under a 32-frame budget. Both methods use CLIP-ViT-L as the frame–query matcher, a 1 fps candidate stream, and identical prompts and evaluation code. Online E2E covers sampling and answer generation but excludes the frame–query matching stage, whose cost both methods share. Sampling GPU memory is the additional footprint beyond the LVLM itself.

Table 3: Component ablation on LongVideoBench (LVB) and MLVU using InternVL-8B with a 32-frame budget. Each variant incrementally adds or changes one component to assess the contribution of adaptive merging and pre-merge graph enhancement.

Variant	**LVB**	**MLVU**
Uniform	58.3	68.3
Top-K Sim.	62.5	73.2
Fine-grained Seg. only	62.8	73.1
+ Adaptive Merge	63.1	74.2
+ Post-merge Graph	63.6	74.5
+ Pre-merge Graph (**Ours**)	**63.9**	**75.1**

InternVL3-8B, with both using CLIP-ViT-L as the matcher, a 1 fps candidate stream, and identical prompts and evaluation code. As shown in Table 2, AGE-Sampler reduces sampling time from 15.880 to 0.426 s/QA ($37.3\times$) and online end-to-end time from 19.101 to 3.987 s/QA ($4.8\times$), while issuing a single LVLM call per question instead of 7.12. Sampling only fits a two-component GMM, propagates over a sparse event graph, and runs a greedy MMR pass over cached features, so its additional GPU footprint is negligible, whereas A.I.R. keeps 16.44–22.72 GiB resident for its iterative reasoning steps. Together with the accuracy in Table 1, this shows AGE-Sampler matches or surpasses A.I.R. at a small fraction of its sampling cost and memory footprint. Our component-wise runtime breakdown is provided in Table 5.

Stable gains across benchmarks. Averaged over the four backbones, Our AGE-Sampler improves uniform sampling by $+6.2\%$ on LongVideoBench, $+8.8\%$ on MLVU, and $+3.8\%$ on VideoMME. The larger gains on MLVU and LongVideoBench, which emphasize long-form and multi-event reasoning, suggest that the improvements mainly come from better event-structure modeling and contextual evidence recovery, rather than merely stronger local query matching.

4.3 Ablation Studies

Component ablation. Following Sec. 3, AGE-Sampler consists of three components: query-gated adaptive-granularity event segmentation, pre-merge graph-enhanced latent event reweighting, and adaptive event-structured frame sampling. We evaluate six variants on InternVL-8B with a 32-frame budget, and report results on LongVideoBench and MLVU in Table 3. Detailed variant settings are provided in Appendix D.

Fine-grained segmentation alone is insufficient. Rows 2 and 3 show that sampling only over fine-grained events performs similarly to top-K similarity sampling. Without merging, the frame budget is scattered across many short, locally relevant segments, leaving insufficient frames for contextual evidence.

Adaptive merging helps but is not sufficient. Rows 3 and 4 show that GMM-based adaptive merging brings clear improvements by preserving fine-grained structures in high-relevance regions while merging low-relevance regions into coarser intervals. However, merging alone cannot recover latent contextual events that receive low direct similarity scores.

Pre-merge graph enhancement beats post-merge. Comparing rows 4 and 5, applying the graph propagation *after* merging yields only a marginal gain, as many latent contextual events have already been absorbed into background segments and can no longer be re-surfaced by propagation. Comparing rows 5 and 6, applying the same propagation *before* merging boosts the importance of latent contextual events while their fine-grained boundaries are still intact, leading to the strongest result and confirming that the order of merging and graph enhancement is itself a meaningful design choice.

Overall, the results demonstrate that adaptive merging and pre-merge graph enhancement are both essential to AGE-Sampler.

Table 4: Performance comparison across Qwen2.5-VL model scales on MLVU under a 32-frame input budget. US denotes uniform sampling, and Gain is the absolute improvement of AGE-Sampler over US.

Model Scale	US	Ours	Gain
Qwen2.5-VL-3B	60.0	67.7	+7.7
Qwen2.5-VL-7B	59.4	69.9	+10.5
Qwen2.5-VL-32B	59.6	72.4	+12.8
Qwen2.5-VL-72B	62.0	**73.9**	+11.9

Scalability across model scales. Table 4 probes how our sampler interacts with backbone capacity, using Qwen2.5-VL at 3B, 7B, 32B, and 72B and reporting MLVU accuracy under a 32-frame budget. Two findings emerge. *(i)* Our method delivers a consistent uplift at every scale ($+7.7\%$ to $+12.8\%$), confirming that the gain does not rely on

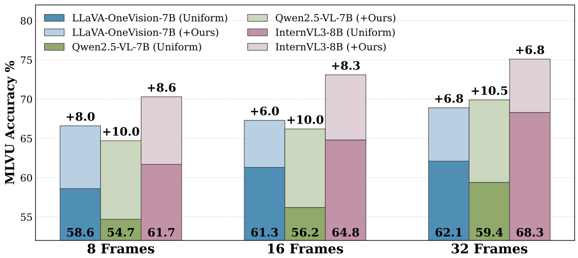

Figure 3: MLVU accuracy under different frame budgets ($K \in \{8, 16, 32\}$) for three LVLM backbones. Our method consistently outperforms the baselines at every budget, and the relative gain over uniform sampling remains stable as K varies.

a particular model size. *(ii)* Uniform sampling (US) plateaus across scales (within $\sim$2 points between 3B and 72B), indicating that frame selection has become the bottleneck; in contrast, the accuracy obtained with our keyframes grows monotonically with scale, peaking at 73.9% on the 72B model. This trend suggests that the keyframes selected by our method better unlock the latent reasoning capacity of larger LVLMs, which is an indirect but informative signal of sampling quality.

Robustness across frame budgets. To probe robustness across frame budgets, we sweep $K \in \{8, 16, 32\}$ over multiple backbones and benchmarks. Figure 3 shows the MLVU result. Our sampler delivers consistent improvements at every budget, and the relative gain over uniform sampling does not collapse when K shrinks from 32 to 8, indicating that the event-level budget allocation remains effective even when the total budget is small. Results on the other two benchmarks follow the same trend and are deferred to the Appendix E.

Additional ablation studies are provided in the Appendix to further examine the robustness and design choices of AGE-Sampler.

5 Conclusion

In this paper, we proposed AGE-Sampler, a training-free adaptive-granularity graph-enhanced event sampling framework for long-video understanding. AGE-Sampler constructs query-adaptive event structures from visual transition and query-relevance signals, and further propagates importance over an event graph to enhance latent contextual events before sampling. This design alleviates the evidence omission problem of pure similarity-based selection while avoiding costly LVLM-assisted frame selection. Experiments on three benchmarks with four LVLM backbones show that AGE-Sampler consistently improves long-video question answering, demonstrating the value of adaptive event modeling and inter-event reasoning.

Limitations

Despite the consistent gains reported above, AGE-Sampler still has several limitations.

Dependence on pretrained matching signals. AGE-Sampler relies on a pretrained image–text matching model to produce the query-relevance signal s_t. Therefore, the quality of event segmentation and graph anchors is bounded by the discriminative ability of this model. When the relevance signal becomes noisy under domain shift or ambiguous queries, the resulting event structure may be suboptimal. Appendix G shows that the method transfers across four different matchers without retuning, retaining at least $+5.2$ points over uniform sampling, so this dependence bounds how good the event structure can become rather than whether the approach works at all.

Limitations of frame-level input compression. Frame-sampling method reduces input cost by selecting a compact subset of frames from a pre-decoded candidate sequence, and is therefore constrained by the coverage of the initial frame decoding stage. If critical evidence is missed during candidate frame decoding, it cannot be recovered by subsequent event modeling or graph propagation. Moreover, frame selection controls the input budget only at the frame level. Future work may explore more robust budget control at richer representation levels, such as visual tokens or spatiotemporal features, to better balance efficiency and information preservation.

Acknowledgments

This work is supported in part by the National Natural Science Foundation of China, under Grant (62676252, 62571298).

References

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, and 8 others. 2025. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*.

Wang Chen, Yuhui Zeng, Yongdong Luo, Tianyu Xie, Luojun Lin, Jiayi Ji, Yan Zhang, and Xiawu Zheng. 2026. Wavelet-based frame selection by detecting semantic boundary for long video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24052–24061.

Bo Fang, YuXin Song, Haoyuan Sun, Qiangqiang Wu, Wenhao Wu, and Antoni Chan. 2026. Threading keyframe with narratives: Mllms as strong long video comprehenders. In *International Conference on Learning Representations*, volume 2026, pages 63417–63431.

Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, Peixian Chen, Yanwei Li, Shaohui Lin, Sirui Zhao, Ke Li, Tong Xu, Xiawu Zheng, Enhong Chen, Caifeng Shan, and 2 others. 2025. Video-MME: The first-ever comprehensive evaluation benchmark of multi-modal LLMs in video analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24108–24118.

Kuofeng Gao, Yang Bai, Jindong Gu, Shu-Tao Xia, Philip Torr, Zhifeng Li, and Wei Liu. 2024. Inducing high energy-latency of large vision-language models with verbose images. In *International Conference on Learning Representations*.

Zhi-Kai Huang and Kwok-Wing Chau. 2008. A new image thresholding method based on gaussian mixture model. *Applied Mathematics and Computation*, 205(2):899–907.

Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. 2025. LLaVA-OneVision: Easy visual task transfer. *Transactions on Machine Learning Research*.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 19730–19742. PMLR.

Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. 2022. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 12888–12900. PMLR.

Shuming Liu, Chen Zhao, Tianqi Xu, and Bernard Ghanem. 2025. BOLT: Boost large vision-language model without training for long-form video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3318–3327.

Yongdong Luo, Wang Chen, Xiawu Zheng, Weizhong Huang, Shukang Yin, Haojia Lin, Chaoyou Fu, Jinfa Huang, Jiayi Ji, Jiebo Luo, and Rongrong Ji. 2025a. QuoTA: Query-oriented token assignment via CoT query decouple for long video comprehension. *ArXiv*, abs/2503.08689.

Yongdong Luo, Xiawu Zheng, Guilin Li, Shukang Yin, Haojia Lin, Chaoyou Fu, Jinfa Huang, Jiayi Ji, Fei Chao, Jiebo Luo, and Rongrong Ji. 2025b. Video-RAG: Visually-aligned retrieval-augmented long video comprehension. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Azra Nasreen and G Shobha. 2013. Key frame extraction using edge change ratio for shot segmentation. *International Journal of Advanced Research in Computer and Communication Engineering*, 2(11):4421–4423.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR.

Kanchana Ranasinghe, Xiang Li, Kumara Kahatapitiya, and Michael Ryoo. 2025. Understanding long videos with multimodal language models. In *International Conference on Learning Representations*, volume 2025, pages 11810–11835.

Xiaoqian Shen, Yunyang Xiong, Changsheng Zhao, Lemeng Wu, Jun Chen, Chenchen Zhu, Zechun Liu, Fanyi Xiao, Bala Varadarajan, Florian Bordes, Zhuang Liu, Hu Xu, Hyunwoo J. Kim, Bilge Soran, Raghuraman Krishnamoorthi, Mohamed Elhoseiny, and Vikas Chandra. 2024. LongVU: Spatiotemporal adaptive compression for long video-language understanding. *ArXiv*, abs/2410.17434.

Yan Shu, Zheng Liu, Peitian Zhang, Minghao Qin, Junjie Zhou, Zhengyang Liang, Tiejun Huang, and Bo Zhao. 2025. Video-XL: Extra-long vision language model for hour-scale video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26160–26169.

Hui Sun, Shiyin Lu, Huanyu Wang, Qing-Guo Chen, Zhao Xu, Weihua Luo, Kaifu Zhang, and Ming Li. 2025. MDP3: A training-free approach for list-wise frame selection in video-LLMs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 24090–24101.

Xi Tang, Jihao Qiu, Lingxi Xie, Yunjie Tian, Jianbin Jiao, and Qixiang Ye. 2025. Adaptive keyframe sampling for long video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 29118–29128.

Xiaohan Wang, Yuhui Zhang, Orr Zohar, and Serena Yeung-Levy. 2024. VideoAgent: Long-form video understanding with large language model as agent. In *European Conference on Computer Vision (ECCV)*, pages 58–76.

Ziyang Wang, Shoubin Yu, Elias Stengel-Eskin, Jaehong Yoon, Feng Cheng, Gedas Bertasius, and Mohit Bansal. 2025. VideoTree: Adaptive tree-based video representation for LLM reasoning on long videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3272–3283.

Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. 2024. LongVideoBench: A benchmark for long-context interleaved video-language understanding. In *Advances in Neural Information Processing Systems*, volume 37, pages 28828–28857. Curran Associates, Inc.

Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, Qianyu Chen, Huarong Zhou, Zhensheng Zou, Haoye Zhang, Shengding Hu, Zhi Zheng, Jie Zhou, Jie Cai, Xu Han, and 4 others. 2024. MiniCPM-V: A GPT-4V level MLLM on your phone. *arXiv preprint arXiv:2408.01800*.

Jinhui Ye, Zihan Wang, Haosen Sun, Keshigeyan Chandrasegaran, Zane Durante, Cristobal Eyzaguirre, Yonatan Bisk, Juan Carlos Niebles, Ehsan Adeli, Li Fei-Fei, Jiajun Wu, and Manling Li. 2025. Rethinking temporal search for long-form video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8579–8591.

Sicheng Yu, Chengkai Jin, Huanyu Wang, Zhenghao Chen, Sheng Jin, Zhongrong Zuo, Xiaolei Xu, Zhenbang Sun, Bingni Zhang, Jiawei Wu, Hao Zhang, and Qianru Sun. 2025. Frame-voyager: Learning to query frames for video large language models. In *International Conference on Learning Representations*, volume 2025, pages 84154–84179.

Yaohua Zha, Yanzi Wang, Hang Guo, Jinpeng Wang, Tao Dai, Bin Chen, Zhihao Ouyang, Xue Yuerong, Ke Chen, and Shu-Tao Xia. 2025. Pma: Towards parameter-efficient point cloud understanding via point mamba adapter. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 16976–16986.

Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11975–11986.

Ce Zhang, Taixi Lu, Md Mohaiminul Islam, Ziyang Wang, Shoubin Yu, Mohit Bansal, and Gedas Bertasius. 2024. A simple LLM framework for long-range video question-answering. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21715–21737, Miami, Florida, USA. Association for Computational Linguistics.

Kaichen Zhang, Bo Li, Peiyuan Zhang, Fanyi Pu, Joshua Adrian Cahyono, Kairui Hu, Shuai Liu, Yuanhan Zhang, Jingkang Yang, Chunyuan Li, and Ziwei Liu. 2025a. LMMs-eval: Reality check on the evaluation of large multimodal models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 881–916, Albuquerque, New Mexico. Association for Computational Linguistics.

Shaojie Zhang, Jiahui Yang, Jianqin Yin, Zhenbo Luo, and Jian Luan. 2025b. Q-Frame: Query-aware frame selection and multi-resolution adaptation for video-LLMs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 22056–22065.

Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Zhengyang Liang, Shitao Xiao, Minghao Qin, Xi Yang, Yongping Xiong, Bo Zhang, Tiejun Huang, and Zheng Liu. 2025. MLVU: Benchmarking multi-task long video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13691–13701.

Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Xuehui Wang, Yue Cao, Yangzhou Liu, Xingguang Wei, Hongjie Zhang, Haomin Wang, Weiye Xu, and 32 others. 2025. InternVL3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*.

Yuanhao Zou, Shengji Jin, Andong Deng, Youpeng Zhao, Jun Wang, and Chen Chen. 2026. A.I.R.: Enabling adaptive, iterative, and reasoning-based frame selection for video question answering. In *International Conference on Learning Representations*, volume 2026, pages 11302–11329.

A Visual Transition Score

The query-agnostic visual transition score $d_t = H(f_{t-1}, f_t)$ of Eq. (1) captures fine-grained shot boundaries that the semantic relevance curve may overlook. We detail the operator $H(\cdot)$ here. Inspired by the content-based detector in PySceneDetect, we compare adjacent frames in a low-level visual feature space. Each frame is converted into visual channels such as hue, saturation, luminance, and optionally edge maps. Let c_t^k denote the k-th visual channel of frame f_t. The transition score is the weighted average of channel-wise mean absolute pixel differences:

$$
\begin{aligned}
H(f_{t-1}, f_t) &= \frac{\sum_{k=1}^{K} w_k\, \delta_t^k}{\sum_{k=1}^{K} |w_k|}, \\
\delta_t^k &= \frac{1}{|\Omega|} \sum_{p\in\Omega} \left| c_t^k(p) - c_{t-1}^k(p) \right|,
\end{aligned} \tag{18}
$$

where Ω is the set of pixel locations, δ_t^k is the mean absolute change of the k-th channel between adjacent frames, and w_k is the weight of the k-th visual-change component. A larger d_t indicates a stronger low-level appearance discontinuity between adjacent frames. The resulting transitions provide structural boundary proposals for the initial fine-grained event set $\mathcal{E}$ in Sec. 3.1; they are query-agnostic and therefore further reweighted and merged in the subsequent stages rather than used as final events.

B Visual Edges in the Event Graph

Following Sec. 3.2, the visual relation A_{ij}^{vis} is composed of a scene-semantic term A_{ij}^{sem} and a local continuity term A_{ij}^{cont}. We provide their detailed definitions here.

Scene-semantic edges. For each event $E_i = [a_i, b_i)$, the event-level visual representation $\mathbf{h}_i$ used in Sec. 3.2 is obtained by average pooling the frame features within its interval:

$$
\mathbf{h}_i = \mathrm{Norm}\left(\frac{1}{b_i - a_i} \sum_{t=a_i}^{b_i-1} \mathbf{x}_t \right), \tag{19}
$$

where $\mathbf{x}_t$ is the visual feature of frame t. We then connect each event to its top-k semantic neighbors whose cosine similarity exceeds a threshold δ:

$$
\begin{aligned}
A_{ij}^{\text{sem}} = \mathbb{I}(j \in \mathcal{N}_k(i))\ \mathbb{I}(\cos(\mathbf{h}_i, \mathbf{h}_j) > \delta) \\
\times \cos(\mathbf{h}_i, \mathbf{h}_j),
\end{aligned} \tag{20}
$$

where $\mathcal{N}_k(i)$ denotes the top-k nearest events to E_i in the visual feature space. This edge captures long-range scene-level associations that cannot be expressed by temporal proximity alone.

Local continuity edges. We further introduce continuity edges between neighboring event candidates to describe whether two adjacent events form a smooth visual transition across their shared boundary. The direction follows the query mode $m(q)$ defined in Sec. 3.2. For $m(q) = \texttt{after}$, the continuity edge points from E_i to E_{i+1}:

$$
A_{i,i+1}^{\text{cont}} = \max\big(0,\ \cos(\mathbf{x}_{b_i-1}, \mathbf{x}_{a_{i+1}})\big)\,. \tag{21}
$$

For $m(q) = \texttt{before}$, the direction is reversed:

$$
A_{i+1,i}^{\text{cont}} = \max\big(0,\ \cos(\mathbf{x}_{b_i-1}, \mathbf{x}_{a_{i+1}})\big)\,. \tag{22}
$$

For $m(q) = \texttt{generic}$, both directions are retained symmetrically. This continuity term helps preserve local process-level transitions between consecutive events when the temporal direction matters.

C Intra-event MMR Term Definitions

In Sec. 3.3, after the event-level budget allocation in Eq. (15), we sample k_i representative frames within each refined event E_i via a localized MMR criterion. The MMR formulation lets us inherit the event-level structure while directly controlling intra-event temporal coverage and redundancy suppression. We provide the detailed definitions of the three terms in Eq. (16) below. Let S_i^{r-1} denote the frames already selected from E_i before the r-th step and $t \in E_i \setminus S_i^{r-1}$ a candidate frame.

Query relevance. The relevance term directly reuses the frame-level semantic signal introduced in Sec. 3.1:

$$
\mathrm{rel}(t) = s_t, \tag{23}
$$

which reflects how well frame f_t matches the query.

Temporal coverage. To encourage the selected frames to cover different temporal positions within the same event, we define the coverage term as the minimum normalized temporal distance between the candidate frame and the already-selected frames:

$$
\mathrm{cov}(t) = \min_{t' \in S_i^{r-1}} \frac{|t - t'|}{|E_i|}, \tag{24}
$$

where $|E_i| = b_i - a_i$ is the temporal length of E_i. A larger $\mathrm{cov}(t)$ means f_t lies in a temporal region not yet covered by S_i^{r-1}, contributing to a more balanced temporal expression within the event. When $S_i^{r-1} = \emptyset$, we set $\mathrm{cov}(t) = 1$.

Visual redundancy. To prevent visually similar frames from being repeatedly selected, we define the redundancy term as the maximum feature similarity between the candidate frame and the already-selected frames:

$$
\mathrm{red}(t) = \max_{t' \in S_i^{r-1}} \cos(\mathbf{x}_t, \mathbf{x}_{t'}). \tag{25}
$$

A larger $\mathrm{red}(t)$ indicates that the candidate is more similar to the existing selections and is therefore more strongly penalized. When $S_i^{r-1} = \emptyset$, we set $\mathrm{red}(t) = 0$.

Greedy selection. Combining the three terms, at the r-th step we greedily pick the candidate with the largest gain,

$$t^* = \arg \max_{t \in E_i \setminus S_i^{r-1}} \left[\lambda_r \operatorname{rel}(t) + \lambda_c \operatorname{cov}(t) - \lambda_d \operatorname{red}(t) \right], \tag{26}$$

and update $S_i^r = S_i^{r-1} \cup \{t^*\}$ until k_i frames have been selected from E_i. The per-event selections are then aggregated into the final keyframe set $Y = \bigcup_{i=1}^{L} Y_i$ with $|Y| = K$.

D Detailed Settings of Component Ablation

Table 3 evaluates six variants on LongVideoBench and MLVU using InternVL-8B under a 32-frame budget. We summarize the detailed settings below.

- **Uniform**: uniformly samples 32 frames from the video, without using query information or event modeling.
- **Top-K Similarity**: selects the top-K frames according to frame-query relevance scores, without event segmentation or graph reasoning.
- **Fine-grained Segmentation only**: performs sampling over fine-grained event segments, without adaptive merging or graph enhancement.
- **Adaptive Merge**: adds GMM-based adaptive merging to refine fine-grained events into adaptive-granularity event intervals, but does not use graph propagation.
- **Post-merge Graph**: applies graph-based importance propagation after adaptive merging, where graph nodes correspond to merged event intervals.
- **Pre-merge Graph (Ours)**: applies graph-based importance propagation before adaptive merging, allowing latent contextual events to be enhanced while their fine-grained boundaries are still preserved.

E Additional Results Across Frame Budgets

In Sec. 4.3, we analyze the robustness of AGE-Sampler under different frame budgets and report the MLVU results in Figure 3. Here, we provide the complementary results on LongVideoBench and Video-MME under the same setting. Specifically, we sweep the frame budget over $K \in \{8, 16, 32\}$ and evaluate multiple LVLM backbones using the same evaluation protocol as in the main experiments.

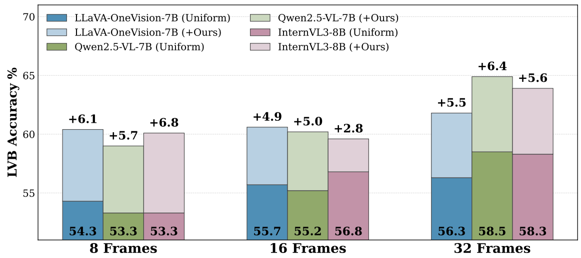

Figure 4: LongVideoBench accuracy under different frame budgets ($K \in \{8, 16, 32\}$) for three LVLM backbones. AGE-Sampler consistently improves over uniform sampling across different budgets, showing that the proposed event-structured frame selection remains effective under both small and moderate frame budgets.

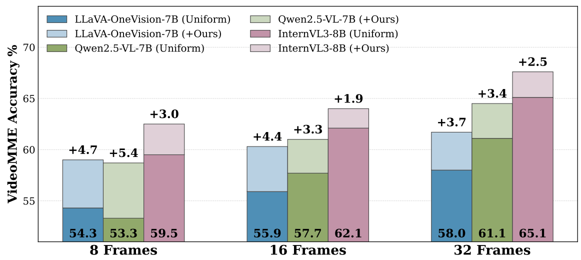

Figure 5: Video-MME accuracy under different frame budgets ($K \in \{8, 16, 32\}$) for three LVLM backbones. AGE-Sampler maintains stable gains across frame budgets, further validating its robustness on benchmarks with diverse video lengths and question types.

As shown in Figures 4 and 5, AGE-Sampler consistently outperforms uniform sampling across different frame budgets. The gains remain stable even when the frame budget is reduced to $K = 8$, indicating that the proposed adaptive-granularity event structure and event-level budget allocation can effectively preserve informative evidence under tight input constraints. Together with the MLVU results in Figure 3, these results demonstrate that AGE-Sampler is robust to the choice of frame budget across multiple long-video understanding benchmarks.

F Efficiency Details

This section reports where the sampling cost of AGE-Sampler is actually spent, breaking the

Stage	Component	Time
Preprocessing	Image–text matching	26.588
	Visual transition	2.646
	Total (s/video)	**29.234**
Sampling	GMM threshold	0.033
	Graph, PPR, merging	0.016
	Budget allocation	$<$0.001
	MMR frame selection	0.359
	Other overhead	0.018
	Total (s/QA)	**0.426**
Evaluation	LVLM generation (s/QA)	3.560

Table 5: Component-wise runtime of AGE-Sampler on VideoMME. Preprocessing is amortized per video and shared across all questions about that video; sampling is per question.

pipeline down into its individual components.

Where the sampling cost goes. Table 5 breaks the pipeline into its three stages. Two points are worth noting. First, the components that constitute the methodological contribution of this work are close to free: the GMM threshold costs 0.033 s/QA and the entire graph stage, covering construction, Personalized PageRank, and merging, costs 0.016 s/QA, together under 12% of sampling time. The dominant term is instead the MMR selection loop at 0.359 s/QA (84.3%), which is a generic greedy procedure and could be reduced further with a batched implementation. Second, preprocessing at 29.234 s/video is the largest single cost, but it is query-independent and amortized over every question asked about that video; it is also incurred by any similarity-driven sampler, which is why the controlled comparison in Table 2 excludes this shared stage.

G Robustness to the Image–Text Matcher

AGE-Sampler derives its query-relevance signal s_t from a pretrained image–text matching model (Eq. (1)), which raises the question of how much of the reported gain is attributable to that matcher rather than to the event-level machinery built on top of it. We therefore replace BLIP-2 (Li et al., 2023) with three alternative matchers of different families and training recipes, namely BLIP-1 (Li et al., 2022), CLIP-ViT-L (Radford et al., 2021), and SigLIP (Zhai et al., 2023), keeping every other component and hyperparameter fixed, and evaluate on MLVU with InternVL3-8B under a 32-frame budget.

As shown in Table 6, accuracy varies by 1.6 points across the four matchers, from 75.1 with BLIP-2 down to 73.5 with SigLIP. The spread is real but small relative to the gain itself: even the weakest matcher improves over the uniform-sampling base of 68.3 by $+5.2$ points, and every variant remains above the top-K similarity baseline of 73.2 reported in Table 3. The ordering is also intuitive, with the two BLIP variants, which are trained with an explicit image–text matching head, outperforming the two contrastive dual-encoder models.

Matcher	MLVU (%)	Δ vs. Uniform
BLIP-2	**75.1**	**+6.8**
BLIP-1	74.6	+6.3
CLIP-ViT-L	73.8	+5.5
SigLIP	73.5	+5.2

Table 6: AGE-Sampler under different image–text matchers on MLVU with InternVL3-8B and a 32-frame budget. All other components and hyperparameters are held fixed. The uniform-sampling base is 68.3. BLIP-2 is the default used throughout the paper.

We draw two conclusions. First, the event-level contribution of AGE-Sampler does not depend on a particular matcher: the method transfers across all four without any re-tuning. Second, matcher quality nonetheless sets a ceiling on the relevance signal, since a better matcher yields a better signal and a correspondingly better event structure. This is the sense in which the dependence noted in the Limitations section is a genuine constraint rather than a fatal one: it bounds how good the segmentation can be, but the graph-based reasoning stage continues to contribute regardless of which matcher supplies the signal.

H Artifact Usage

We use publicly available benchmarks, LVLM backbones, pretrained image–text matching models, and baseline implementations only for research and evaluation purposes. For each artifact, we follow its publicly released license, terms of use, and access conditions. The datasets are used only for benchmark evaluation, and the pretrained models and baseline implementations are used only to reproduce or compare frame selection methods under matched settings.

I LLM Usage

We used an OpenAI LLM, specifically GPT-5.5, as a writing assistant. The model was used to refine grammar and phrasing, improve clarity, and suggest edits to figure and table captions. It did not contribute to research ideation, experimental design, implementation, data analysis, or technical content beyond surface-level writing assistance. All LLM-generated suggestions were carefully reviewed and edited by the authors, who take full responsibility for the final text, figures, and tables.